

Java Backend Developer in 30 Days

A daily plan: Java → Maven → Spring Boot → REST APIs → Security → Docker → Deploy.
Each day: focused tasks + a short YouTube video (≤ 25 min) to watch and code along.

- 30 DaysFull roadmap
- 4 WeeksClear phases
- 5+ ProjectsBuild & test
- Spring BootCore framework
- DeployDocker + CI/CD

Windows Dev Setup — Java & IntelliJ

Install once, use all month

- 1

Install JDK 21 LTS
adoptium.net → Windows x64 MSI → run installer → adds JAVA_HOME automatically
[adoptium.net](#)
- 2

Verify Java install
Open Command Prompt: java -version → should show openjdk 21
- 3

Install IntelliJ IDEA
jetbrains.com/idea → Community Edition (free) → download .exe → install
[jetbrains.com/idea](#)
- 4

Install Maven
maven.apache.org → download zip → extract to C:\maven → add bin to PATH
[maven.apache.org](#)
- 5

Install Postman
postman.com → download for Windows → use to test REST endpoints
[postman.com](#)
- 6

Install PostgreSQL
postgresql.org → download installer → set password: postgres → port 5432
[postgresql.org](#)
- 7

Spring Initializr
start.spring.io → Maven, Java 21, add: Web, JPA, PostgreSQL → Generate zip
[start.spring.io](#)
- 8

IntelliJ tips
Ctrl+Shift+F10 Run · Ctrl+Alt+L Format · Alt+Insert Generate code

WEEK 1 —Java Foundations & OOP			Days 1–7
Day 1 Setup Java Setup & First Program - Install JDK 21 LTS from adoptium.net - Install IntelliJ IDEA Community (free) - Write and run Hello World — understand structure  Amigoscode · Java Intro · ~15min	Day 2 Java Java Variables & Types - Primitive types: int, double, boolean, char - String operations: length, concat, substring - Type casting and operators: +, -, *, /, %  Telusko · Java Variables · ~15min	Day 3 Java Control Flow - if / else if / else and switch - for, while, do-while loops - break, continue and nested loops  Amigoscode · Java Control Flow · ~18min	
Day 4 Java Methods & Arrays - Define and call methods, return values - Method overloading — same name, diff params - Arrays: declare, initialize, loop through  Traversy · Java Arrays · ~20min	Day 5 OOP Object-Oriented: Classes - Class, object, constructor, fields - this keyword and access modifiers - Create a Student class with name, grade fields  Amigoscode · OOP Classes · ~20min	Day 6 OOP OOP: Inheritance & Interfaces - extends: child inherits parent fields/methods @Override to replace parent behavior - implements: define a contract with interface  Telusko · Java Inheritance · ~18min	
Day 7 Project Week 1 Project: Library Console App - Build a Book class with title, author, year - Store books in an ArrayList - Add methods: add, remove, search, print all  Amigoscode · Java Project · ~20min			

WEEK 2 —Maven, REST Concepts & Spring Boot			Days 8–14
Day 8 Maven Maven Basics - What Maven does: build, dependencies, plugins - pom.xml structure: groupId, artifactId, version - Add a dependency (Jackson) — mvn install  JavaGuides · Maven Crash · ~20min	Day 9 API REST API Concepts - HTTP verbs: GET, POST, PUT, DELETE - JSON format: objects, arrays, key-value - Status codes: 200, 201, 400, 404, 500  Traversy · REST API Concepts · ~20min	Day 10 Spring Spring Boot First App - Go to start.spring.io — generate project - Open in IntelliJ — run the app - Understand @SpringBootApplication entry point  Amigoscode · Spring Boot Intro · ~20min	
Day 11 Spring Spring Boot REST Controller @RestController and @GetMapping - Return JSON string and Java objects - Test endpoint in browser / Postman  Amigoscode · REST Controller · ~20min	Day 12 DB Spring Boot + PostgreSQL - Add Spring Data JPA + PostgreSQL driver - Configure application.properties: url, user, pass @Entity, @Id, @GeneratedValue annotations  Amigoscode · Spring + Postgres · ~22min	Day 13 Spring CRUD with JPA Repository - JpaRepository: findAll, save, findById, delete @PostMapping to create, @DeleteMapping to delete - Return proper HTTP status codes in responses  JavaGuides · Spring CRUD · ~20min	
Day 14 Project Week 2 Project: Student REST API - Entity: Student (id, name, email, grade) - Endpoints: GET all, GET by id, POST, DELETE - Test all endpoints with Postman  Amigoscode · Spring Boot Demo · ~20min			

WEEK 3

—Spring Depth: Security & Testing

Days 15–21

Day 15 Spring Service Layer & DTOs @Service class between controller and repo - DTO: Data Transfer Object pattern explained - Map entity to DTO — hide internal fields  Java Guides · Service Layer · ~18min	Day 16 Spring Exception Handling @ExceptionHandler and @ControllerAdvice - Custom exception classes: ResourceNotFoundException - Return structured error JSON: code, message  Java Brains · Exception Handling · ~20min	Day 17 Spring Request Validation - Add spring-boot-starter-validation dependency @NotNull, @NotBlank, @Size, @Email annotations @Valid on @RequestBody — return 400 on errors  Amigoscode · Validation · ~15min
Day 18 Security Spring Security Basics - Add spring-boot-starter-security - Basic Auth: in-memory user config - Permit public routes, protect private ones  Amigoscode · Spring Security · ~20min	Day 19 Security JWT Authentication - What is JWT: header, payload, signature - Generate token on login, verify on each request - Return 401 Unauthorized for invalid tokens  Amigoscode · JWT Tutorial · ~22min	Day 20 Testing Unit Testing with JUnit 5 @Test, @BeforeEach, @AfterEach annotations - assertEquals, assertNotNull, assertThrows - Test the Service layer — mock with Mockito  Amigoscode · JUnit Tutorial · ~20min
Day 21 Project Week 3 Project: Secure API - Add JWT auth to Student REST API - Protect POST and DELETE with token - Write 5 unit tests for the service layer  Amigoscode · Spring Security Project · ~20..		

WEEK 4

—Production: Deploy, CI/CD & Docs

Days 22–30

Day 22 Config Spring Profiles & Config - application-dev.yml vs application-prod.yml @Value and @ConfigurationProperties - Switch active profile with env variable  Java Guides · Spring Profiles · ~15min	Day 23 Spring Pagination & Sorting - Pageable parameter in repository method - Return Page<T>: content, totalPages, number - Sort by field name ASC / DESC via query param  Java Brains · Pagination · ~18min	Day 24 Spring File Upload & Storage - MultipartFile in @PostMapping endpoint - Save file to local disk or GCS/S3 bucket - Return file URL in response body  Amigoscode · File Upload · ~20min
Day 25 Ops Logging with SLF4J & Logback - Logger logger = LoggerFactory.getLogger(...) - log.info(), log.warn(), log.error() levels - Configure logback-spring.xml for output format  Telusko · Logging · ~15min	Day 26 Deploy Docker Basics for Spring Boot - Write a Dockerfile: FROM, COPY, EXPOSE, CMD - docker build -t myapp . then docker run - Understand containers vs virtual machines  Amigoscode · Docker + Spring · ~20min	Day 27 CI/CD CI/CD with GitHub Actions - Create .github/workflows/build.yml - Steps: checkout, setup-java, mvn test, build - Trigger on push to main branch  Amigoscode · GitHub Actions · ~20min
Day 28 Deploy Cloud Deploy: Railway / Render - Push Docker image to Docker Hub - Connect GitHub repo to Railway.app - Set env vars for DB_URL, JWT_SECRET  Traversy · Deploy Spring · ~20min	Day 29 Docs API Documentation with Swagger - Add springdoc-openapi dependency - Access /swagger-ui.html in browser - Annotate endpoints with @Operation, @ApiResponse  Java Guides · Swagger Docs · ~15min	Day 30 Project Final Project: Complete Backend API - Pick domain: Tasks / Products / Users - Full CRUD + Auth + Validation + Tests - Deploy live: share Swagger docs URL  Amigoscode · Spring Boot Full Project · ~2..

Completion goal:

Deploy your Spring Boot API with JWT auth on Railway. Write unit tests and share your Swagger docs URL in your portfolio.